

10-Mark Questions & Answers — Data Structures Module 2: Stacks & Queues

Q1. Explain Stack & Queue ADTs, Infix to Postfix Conversion & Evaluation, Circular Queue, and Deque in detail with algorithms and executable C programs.

Comprehensive 10-mark master answer covering:

1. Stack ADT array implementation with push/pop/peek functions.
2. Complete step-by-step trace and C program for Infix to Postfix conversion.
3. Postfix Expression Evaluation algorithm using Stack.
4. Circular Queue array implementation with modulo index arithmetic.
5. Double-Ended Queue (Input-Restricted and Output-Restricted Deques) and applications.